Dear CPSE Master's Students,

I am reaching out to you following the recommendation of your instructor, Mrs. Yahia, who is confident that you can assist us in deploying a supervision system within our Dyneo learning factory, in order to transform it into a 4.0 learning factory.

Let me explain everything! The main mission of the Dyneo learning factory is to contribute to the deployment of operational excellence as a driver of competitiveness. This approach relies on the identification of malfunctions, waste, and organizational or communication problems that often lead to significant losses in companies.

During the training, learners assemble snowshoes across several workstations. Production monitoring is currently carried out through operators or observers who record data such as cycle times per station, actual takt time, defects, and quality issues. Decision-making regarding corrective actions is based on these displayed indicators , and sometimes, decisions are made without sufficient justification.

Our goal is to integrate new Industry 4.0 technologies to demonstrate how they can facilitate the implementation of Lean approaches.

We are therefore interested in real-time supervision of the production system, seen as a cyber-physical production system, and would like to seek your assistance , as several development projects could be considered.

We invite you to work on the use case **"Ensuring better responsiveness in incident resolution"**, described in the document attached to my email.

The objective is to detect and qualify a problematic situation, and to define a decision-support system capable of diagnosing, specifying the necessary actions, and, why not, predicting issues in order to reduce reaction time!

You are expected to adopt a systems engineering approach in designing the demonstrator and to address the following needs:

1. Can you help us specify the functional requirements (functional specification document) to accelerate decision-making for managers?
2. Can you define the technical architecture of the proposed supervision system?
    a. Which IoT devices would you recommend?
    b. Which communication protocols should be prioritized?
    c. What would be the proposed architecture for the decision-support system?
3. The IT department (DSI) mentioned the use of Node-RED, which seems to be easy to handle. Could you use it to deploy a Proof of Concept (PoC)?
4. Finally, we would like to specify a system to evaluate ROI (Return on Investment) following the integration of these new technologies.

Mrs. Yahia, who is well aware of our needs, will guide you if you encounter any blocking points.

Thank you in advance for your contribution. We look forward to receiving your deliverables soon.

Kind regards,

The DYNEO Team

Use Case: *Ensuring better responsiveness in incident resolution*

Current situation

We currently operate a flexible assembly line composed of four workstations, where each operator performs clearly defined tasks.

The production manager monitors operations through a performance board that indicates whether the line is on target, ahead, or delayed relative to the takt time. This board is updated by the last operator once each product assembly is completed.

When a deviation occurs, the manager must investigate the root causes directly on the shop floor, without interrupting the assembly process. This investigation involves moving from workstation to workstation to identify the bottleneck, often by visually observing the accumulation of parts at each station's input.

To gain a deeper understanding of the issue, the manager also inspects the red bins containing reworked or defective parts to identify recurring quality problems. Based on these observations, the problematic workstation can usually be identified; however, determining the exact root cause remains challenging.

Several potential factors may explain the deviation:

- **Human factors:** lack of operator training, fatigue, high cognitive load, stress, or unfavorable environmental conditions (noise, temperature), all of which can slow down task execution.
- **Process issues:** disruptions or delays originating from upstream workstations.
- **Material issues:** defective or inconsistent parts affecting downstream operations.
- **Environmental issues:** High or low temperatures, excessive noise, or other unfavorable environmental conditions.

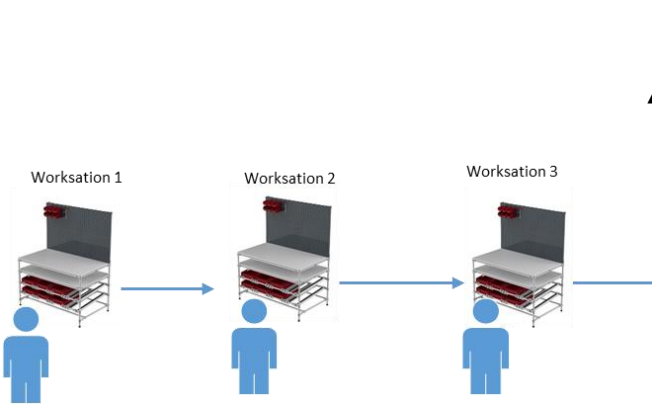
While deviations are detected quickly via the performance board, understanding their root cause in real time is difficult. This manual and reactive investigation process is time-consuming and often delays the implementation of corrective actions.

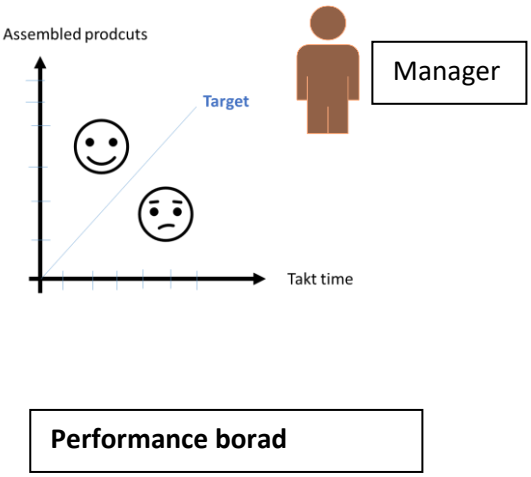
Target situation

The manager needs enhanced visibility and decision-support tools to improve responsiveness and autonomy in incident resolution.

- Detect and characterize anomalies in real time, beyond simple deviation indicators.
- Identify and qualify root causes quickly, using contextual and historical data.
- Recommend corrective actions from a predefined problem–action database when the issue is known.
- Initiate a causal analysis process based on collected data when a new or unknown problem arises.

- Anticipate potential problems before they occur by analyzing early warning signals and abnormal trends.**<u>Assembly line and performance dashboard illustration</u>**





Performance borad

